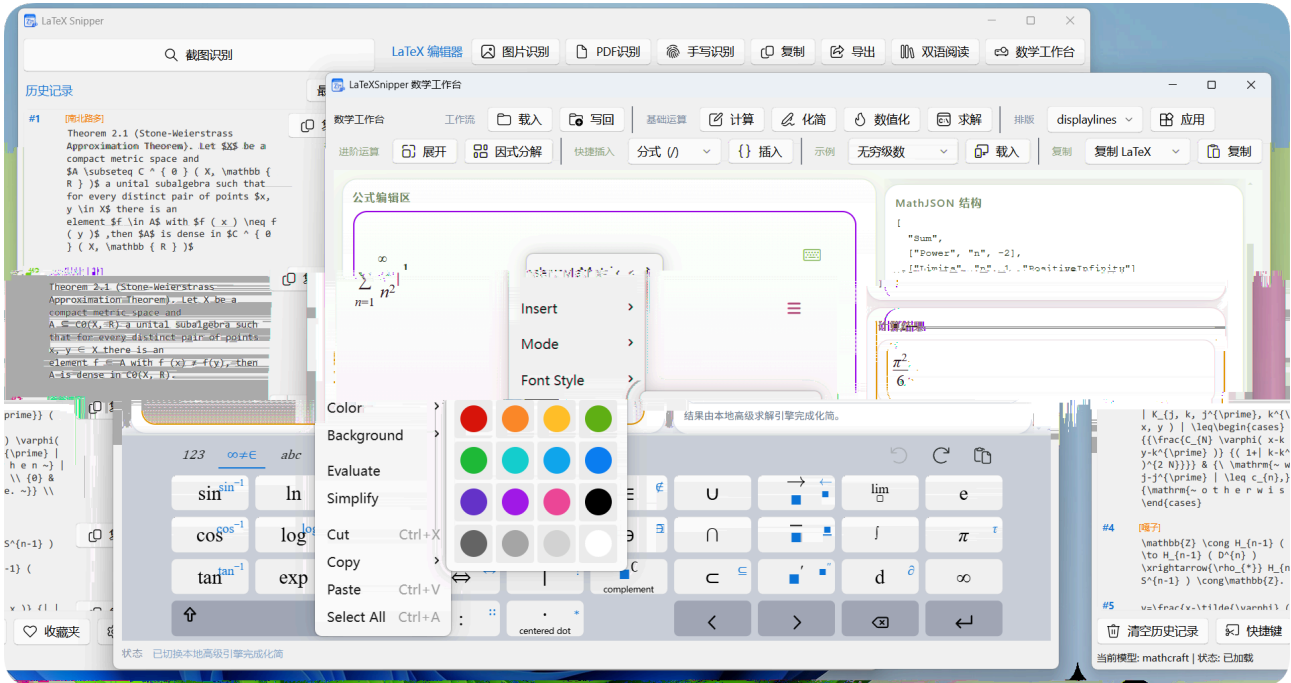


v2.3.2_Final_Stable |



-
- MathCraft ONNX
-
-
-
-
- PDF
- Typst
-
- Linux / macOS
-
-
- Bug
- MathCraft OCR
-

LaTeXSnippet

Python

OCR



Python

apt/brew

- Windows
- Linux python3 -m venv Debian/Ubuntu python3-venv
- macOS Python 3.10+ Homebrew Python python.org

- 1.
2. LaTeX
3. LaTeX
4. “ ”
- 5.
6. 30+ Pandoc

- %USERPROFILE%\latexsnipper\logs\ %LOCALAPPDATA%\LaTeXSnippet\logs\
- ~/.latexsnipper/logs/
- ~/.latexsnipper/logs/
- crash-native.log



Bug

logs



```
warmup()      warmup
python -m mathcraft_ocr models check
```

“ model xxx downloading ”

- [GitHub Releases](#)
- [/](#)
- `urllib`
- [GitHub](#)
- `HTTP_PROXY / HTTPS_PROXY`
- [MathCraft](#)
 - Windows `%APPDATA%\MathCraft\models\`
 - Linux/macOS `~/.mathcraft/models/`



```
python -m mathcraft_ocr models check      python -m mathcraft_ocr
models download
```

`failed to import onnxruntime` `missing get_available_providers`

- `onnxruntime`
- `pip`
- CPU GPU
- `pip uninstall onnxruntime onnxruntime-gpu`
- GPU `pip install onnxruntime-gpu`
- CPU `pip install onnxruntime`

`warmup` `CUDA` `cuda wasn't able to be loaded` `cublas`

- `onnxruntime-gpu` `CUDA / cuDNN`
- `CUDA` `onnxruntime-gpu`
-
- GPU

- MATHCRAFT_FORCE_ORT_CPU=1 CPU
- CUDA Toolkit cuDNN onnxruntime-gpu onnxruntime
-

warmup invalid protobuf failed to load model sha256 mismatch

- Windows %APPDATA%\MathCraft\models\<model_id>\ Linux/macOS
~/.mathcraft/models/<model_id>/
-
- python -m mathcraft_ocr models check

warmup CUDA out of memory
GPU

- GPU
- MATHCRAFT_FORCE_ORT_CPU=1 CPU

X

“ ”



API Key	Base URL	LaTeXSnipper	API Key
---------	----------	--------------	---------

- 404409
- MinerU
- /health
- /file_parse
- MinerU
 - “http://127.0.0.1:8000/health”
 - MinerU 409MinerU

- /v1/models404
- Ollama
- /api/tags
- /v1/models
- “OpenAI-compatible”
- “Ollama”

- Ollama →
 - / DeepSeek / OpenAI / Groq →
 - MinerU →
 - MathCraft ONNX →

LaTeX

“ LaTeX ”

Python

- Python Python 3.11
- Python 3.10+
- Python 3.11 Windows

```
# Python 3.11 创建虚拟环境
python3.11 -m venv .venv

# Windows 激活
.\.venv\Scripts\activate

# Linux/macOS 激活
source .venv/bin/activate

# 安装依赖
pip install -r requirements.txt
```

GitHub Release

exe

git clone

Windows

Python 3.11

Linux/

macOS

PyInstaller

Python

- Release
- requirements.txt requirements-build.txt

Linux/macOS

src/deps/python311

MathCraft Pandoc Python

```
~/ .latexsnipper/deps/python311
```

```
Linux/macOS /usr/lib/latexsnipper .app
~/ .latexsnipper/deps/python311 venv
pyvenv.cfg CI
```

Linux .deb

python3

python3-venv

macOS

.deb

python3

```
# Homebrew
brew install python
```

或使用 python.org 官方 macOS 安装包
<https://www.python.org/downloads/macos/>

```
pip install -r requirements.txt
```

- pywin32 wheel
- PyQt6
- PyPI
- / PyPI

```
pip install -i https://pypi.tuna.tsinghua.edu.cn/simple -r requirements.txt
```

- pywin32 pip PyPI wheel

```
C:\Program Files\  
%APPDATA% ~/.latexsnipper/  
UAC
```

-
- %LOCALAPPDATA%

Windows	%APPDATA%	ASCII	C
ONNX	MATHCRAFT_HOME	ASCII	C:\mathcraft_data

Ollama / MinerU
Windows Defender

LaTeXSnipper

- Python / Ollama
-
- Ollama 0.0.0.0 127.0.0.1

api.github.com

-
- GitHub Releases
- %USERPROFILE%\latexsnipper/logs/

exe

PyInstaller exe

-
- SignPath

PDF

- DPI < 100 > 200
- PDF
-

- PDF 140-170 DPI
- 200-300 DPI
- “ ”

LaTeX

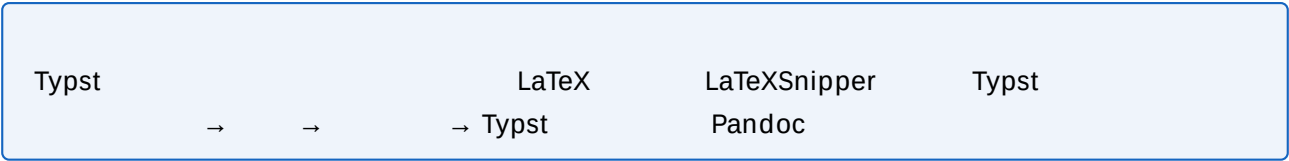
Markdown
MinerU

4K

Base64

33%

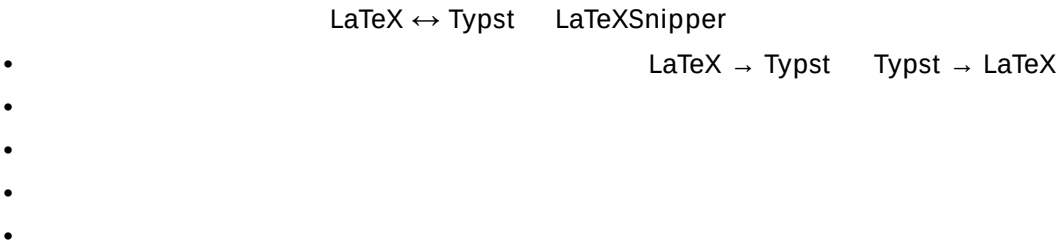
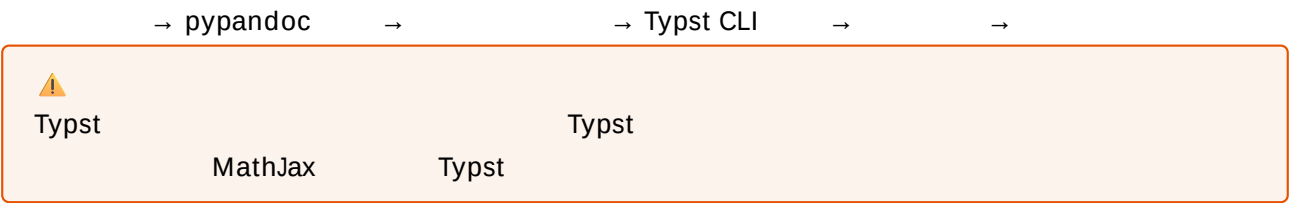
-
-
-



LaTeXSnipper



Typst



“ Typst ”

LaTeX/Typst

- Typst CLI PATH
- Typst
- Typst --format svg
- pandoc LaTeX Typst

- Typst

- <https://github.com/typst/typst>

```
# Windows (scoop / winget)
winget install Typst.Typst

# macOS
brew install typst

# Linux
cargo install typst-cli
```

- “ ” → → “ Typst ” → “ ” Typst CLI →

- LaTeX Typst

```
pip install pandoc
```

Pandoc

<https://pandoc.org>

- Typst CLI `typst --version` ≥ 0.11

“ ”

- Typst

-

- Typst

- `typst --version`

- Windows `C:\Users\<用户名>\.cargo\bin\typst.exe` `cargo` `C:\Program Files\Typst\typst.exe` `winget`

- macOS/Linux `/usr/local/bin/typst` `~/.cargo/bin/typst`

- `PATH` `typst`

Typst

- `py pandoc` LaTeX → Typst LaTeX

- Typst LaTeX $\mathbb{R}$ vs $\mathbb{R}$

- Typst LaTeX

- “ ” “ MathJax ” “ LaTeX + pdf latex/xelatex ”

- Typst Typst

- [Typst] Pandoc conversion [ERROR] Typst compile failed

`py pandoc` `latex → typst`

LaTeX

- `\newcommand`

- AMSmath `\substack` `\xrightarrow`

- [/](#)
- LaTeX

MathJax
 LaTeX

Typst 15

- MathJax
- /CPU
- Typst

Pandoc .typ Typst CLI

- Pandoc typst Typst
- Pandoc LaTeX
- Typst Pandoc

- Pandoc ≥ 3.0 `pandoc --version`
- Typst ≥ 0.11
- .typ
- LaTeX

LaTeXSnipper MathCraft OCR Markdown typst_formulas

True LaTeX Typst

- `py pandoc` `pip install py pandoc`
- Pandoc
- `py pandoc` LaTeX



Python

```
python -c "from core.mathcraft_document_engine import convert_latex_to_typst;
print(convert_latex_to_typst(r'\\int_{0}^{\\infty} e^{-x} dx'))"
```

•

• Wayland

•

Word / EPUB / Typst
Pandoc Pandoc

• “ PANDOC ”
• Pandoc <https://pandoc.org> PATH
• LaTeX / Markdown / MathML / HTML Pandoc

• OCR

•

•

• “ ”
• “ ”
• “ ”

• “ ”
• “ ”

• 700ms > 3 300ms —
• 250ms ≤ 2 —
• 350ms —

• + OCR
• UnsharpMask
• QImage → PIL PNG
• < 120px 2x

- *
- Compute Engine

- $2x$ $2*x$
- SymPy/mpmath
-

[LaTeX]

- Typst Typst
- LaTeX
- v2.3.2 LaTeX Typst
- [LaTeX] LaTeX Typst

-
- “ ” “ ” “ ”

.docx .epub
Pandoc UTF-8

- Pandoc ≥ 3.0
- LaTeX PDF ctex
- Word Word

Wayland
Wayland

Qt

- grim + slurp wlrroots gnome-screenshot GNOME
-
-

Python grim main gnome-screenshot
—— Qt

EGL not available No physical devices GL0zone not found Failed to get system
egl display

MathCraft GPU Qt WebEngine / Chromium Wayland
WSL DRI render EGL/GPU
LaTeXSnippet Linux X11 + GPU +
/dev/dri/renderD*

```
# 强制启用 Linux 图形兜底
LATEXSNIPPER_FORCE_LINUX_GRAPHICS_FALLBACKS=1 latexsnippet

# 禁用 Linux 图形兜底, 尝试原生 Qt/GPU 路径
LATEXSNIPPER_DISABLE_LINUX_GRAPHICS_FALLBACKS=1 latexsnippet
```

XWayland/XCB/QtWebEngine Debian/Ubuntu

```
sudo apt install xwayland libxcb-cursor0 libxcb-icccm4 libxcb-image0 \
libxcb-keysyms1 libxcb-render-util0 libxcb-xinerama0 libxkbcommon-x11-0
```

macOS “ ” “ ”
Apple notarization

- → → “ ”
- xattr -cr /Applications/LaTeXSnippet.app

- → → LaTeXSnippet

- | | pip | pip / setuptools / wheel |
|---------------------------------|--------|--------------------------|
| macOS | Python | Python 3.10+ |
| ~/ .latexsnipper/deps/python311 | Python | venv / ensurepip / pip |

- Homebrew Python `brew install python`
- python.org macOS Python
- LaTeXSnippet

Windows: %USERPROFILE%\latexsnipper\
Linux: ~/.latexsnipper/
macOS: ~/.latexsnipper/

```
Linux:    ~/.latexsnipper/deps/python311
macOS:   ~/.latexsnipper/deps/python311
```

```
Windows: %APPDATA%\MathCraft\models\
Linux:   ~/.mathcraft/models/
macOS:   ~/.mathcraft/models/
```

LaTeXSnippet

- LaTeXSnippet
- Python
- Explorer
- open Finder
- xdg-open x-terminal-emulator
- CMD
- Terminal.app

GUI

Python

PATH

Python

- where python Windows / which python Linux/macOS

-

-

Python

hook

LaTeXSnipper

X

=

=

Bug

%USERPROFILE%\latexsnipper\logs\ %LOCALAPPDATA%\LaTeXSnipper\logs\
stderr

crash-native.log

exe

Python

- <https://github.com/SakuraMathcraft/LaTeXSnipper/issues>
- <https://github.com/SakuraMathcraft/LaTeXSnipper/discussions>

README



- `python -m venv .venv → pip install -r requirements.txt → python src/main.py`
- `python3 -m venv .venv → pip install -r requirements-linux.txt → python src/main.py`
- `python3 -m venv .venv → pip install -r requirements-macos.txt → python src/main.py`
- `requirements-linux.txt requirements-macos.txt -r requirements.txt`
`pynput`

`python311/` `src/deps/python311/` `Python`

- `python311/` `Windows` `Python`
`ruff/pyright/pytest`
- `src/deps/python311/` `IDE` `Actions` `requirements*.txt`
`ruff pytest pyright`
- `Linux/macOS` `src/deps/python311` `~/.latexsnipper/`
`deps/python311`

```
py -3.11 -m venv src\deps\python311
.\src\deps\python311\python.exe -m pip install --upgrade pip wheel setuptools
.\src\deps\python311\python.exe -m pip install -r requirements.txt
.\src\deps\python311\python.exe -m pip install -r requirements-build.txt
.\src\deps\python311\python.exe -m pip install ruff pytest pyright
```

- `python311/ — Windows`
- `src/deps/python311/ —`
- `~/.latexsnipper/deps/python311 — Linux/macOS`

`docs/developer_code_standards.md`

```
.\src\deps\python311\python.exe -m ruff check .
.\src\deps\python311\python.exe -m pytest test
.\src\deps\python311\python.exe -m pyright
.\src\deps\python311\python.exe -m compileall -q -x "src[\\/] +deps" src mathcraft_ocr test
```



`ruff pytest pyright` `requirements*.txt`

```
.\src\deps\python311\python.exe -m pip install ruff pytest pyright
```

pyrightconfig.json

src/deps/python311

python311/

Windows

pyright

	src/deps/python311	python311
---	--------------------	-----------

MathCraft OCR

LaTeXSnippet

ONNX

OCR

- GitHub <https://github.com/SakuraMathcraft/MathCraft-Models>
- PyPI <https://pypi.org/project/mathcraft-ocr/>
-
- ONNX Runtime CPU CUDA GPU
- PDF
-

```
pip install mathcraft-ocr
```

LaTeXSnippet

```
# 检查模型缓存状态
python -m mathcraft_ocr models check

# 手动下载模型
python -m mathcraft_ocr models download

# 识别图片中的公式
python -m mathcraft_ocr run --image formula.png
```

“ MathCraft ONNX ”

CPU

GPU

mathcraft-ocr PyPI

0.1.8

MathCraft Models v1.0.0

- mathcraft-formula-det — Formula Detection
- mathcraft-formula-rec — LaTeX Formula Recognition
- mathcraft-text-det — Text Detection
- mathcraft-text-rec — Text Recognition



- formula — → LaTeX
- text — OCR →
- mixed — + → Markdown

CHAPTER 1. PRELIMINARIES

0

PROOF. (1) If $A \cap B = \emptyset$, then the theorem follows immediately since both $A \cup B$ and $A' \cap B'$ are empty sets. Otherwise, we must show that $(A \cup B) \subseteq A' \cap B'$ and $(A \cup B) \supseteq A' \cap B'$. Let $x \in A \cup B$. Then $x \in A$ or $x \in B$. So $x \in A'$ and $x \in B'$ by definition of the union of sets. By the definition of the complement, $x \in A'$ and $x \in B'$, therefore, $x \in A' \cap B'$ and we have $(A \cup B) \subseteq A' \cap B'$.

To show the reverse inclusion, suppose that $x \in A' \cap B'$. Then $x \in A'$ and $x \in B'$, and $x \notin A$ and $x \notin B$. Hence, $x \in A \cup B$. So $(A' \cap B') \subseteq A \cup B$. The proof of (2) is left as an exercise.

Example 1.4 Other relations between sets often hold true. For example,

$$(A \setminus B) \cap (B \setminus A) = \emptyset$$

To see that this is true, observe that

$$\begin{aligned} (A \setminus B) \cap (B \setminus A) &= (A \cap B') \cap (B \cap A') \\ &= A \cap A' \cap B \cap B' \\ &= \emptyset. \end{aligned}$$

Cartesian Products and Mappings

Given sets A and B , we define a new set $A \times B$ called the *Cartesian product* of A and B as the set of ordered pairs. That is,

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

Example 1.5 If $A = \{1, 2, 3\}$ and $B = \{1, 2, 3\}$, then $A \times B$ is the set

$$\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$$

and

$$A \times \emptyset = \emptyset.$$

we define the *Cartesian product* of $A_1, A_2, \dots, A_n$ as

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, \dots, a_n) : a_i \in A_i \text{ for } i = 1, \dots, n\}.$$

If $A_1 = A_2 = \dots = A_n = A$, we often write A^n for $A_1 \times A_2 \times \dots \times A_n$ (where A is written n times). For example, the set $\mathbb{R}^3$ consists of all of 3-tuples of real numbers.

Subsets of $A \times B$ are called *relations*. We will define a *mapping* or *function* from a set A to a set B as a special type of relation where each element

a unique element $a \in A$ is associated with a unique element $b \in B$. Another way of saying this is that for every element a in A , there is a unique element b in B such that $(a, b) \in f$. We usually write $f : A \rightarrow B$ or $A \rightarrow B$ instead of writing down ordered pairs (a, b) . We write $f(a)$ for the element b such that $(a, b) \in f$. The set A is called the *domain* of f and the set B is called the *range* or *image* of f .

$$f(A) = \{f(a) : a \in A\} \subseteq B$$

we can think of the elements in the function's domain as input values and the elements in the function's range as output values.

where

$$\begin{aligned} F_{11}(u_n, v_n, \delta) &= -\frac{(1+b)x^*\delta}{2a} - \frac{a \sin x^* u_n^2}{2} - \frac{(1+b)x^*\delta^2}{4a^2} \\ &\quad - \frac{a \cos x^* u_n^3}{6} - \frac{(1+b)x^*\delta^3}{8a^3} + O(\|(u_n, v_n, \delta)\|^4), \\ F_{12}(u_n, v_n, \delta) &= -\frac{(-x^*b^2 + 2a \sin x^* - 2x^*b - x^*)\delta}{2a} \\ &\quad + \frac{a(b-1) \sin x^* u_n^2}{2} + \frac{(1+b)^2 x^* \delta^2}{4a^2} - \cos x^* \delta u_n \\ &\quad + \frac{a(b-1) \cos x^* u_n^3}{6} + \frac{(1+b)^2 x^* \delta^3}{8a^3} + \frac{\sin x^* \delta u_n^2}{2} \\ &\quad + O(\|(u_n, v_n, \delta)\|^4). \end{aligned}$$

Consider the following system

$$\begin{pmatrix} u_{n+1} \\ v_{n+1} \\ \delta_{n+1} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_n \\ v_n \\ \delta_n \end{pmatrix} + \begin{pmatrix} \hat{F}_{11}(u_n, v_n, \delta_n) \\ \hat{F}_{12}(u_n, v_n, \delta_n) \\ 0 \end{pmatrix}, \quad (15)$$

where

$$a_{11} = -1 + a \cos x^*, \quad a_{12} = 1,$$

then the system (15) becomes

$$\begin{pmatrix} x_{n+1} \\ y_{n+1} \\ \tilde{\delta}_{n+1} \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -b + a \cos x^* & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tilde{x}_n \\ \tilde{y}_n \\ \tilde{\delta}_n \end{pmatrix} + \begin{pmatrix} \tilde{F}_{11}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\ \tilde{F}_{12}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\ 0 \end{pmatrix},$$

(16)

where

$$\begin{aligned} \tilde{F}_{11}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) &= -\frac{\tan x^* \sec x^* \tilde{x}_n^2}{2a} - \frac{\tan x^* \tilde{x}_n \tilde{y}_n}{b-1} + \frac{\sin x^* \tan x^* \tilde{\delta}_n \tilde{x}_n}{2 \cos x^* (a-2b-2)} \\ &\quad - \frac{a \sin x^* \tilde{y}_n^2}{2(b-1)^2} + \frac{a \sin^2 x^* \tilde{y}_n \tilde{\delta}_n}{2(b-1)(a \cos x^* - b - 1)} \\ &\quad - \frac{\tilde{\delta}_n^2}{8(a \cos x^* - b - 1)^2 a^2} \left((2x^*(b+1)^3 + a^3 \sin^3 x^* \right. \\ &\quad \left. - 4ax^*(b+1)^2 \cos x^* + 2a^2 x^*(b+1) \cos^2 x^*) \right) \\ &\quad + \frac{\sec^2 x^* \tilde{x}_n^3}{6a^2} + \frac{\sec x^* \tilde{x}_n^2 \tilde{y}_n}{2a(b-1)} - \frac{\tan x^* \tilde{x}_n^2 \tilde{\delta}_n}{4a(a \cos x^* - b - 1)} \end{aligned}$$



第十一届清疏大学生数学竞赛班讲义

非数学类 & 数学类共用

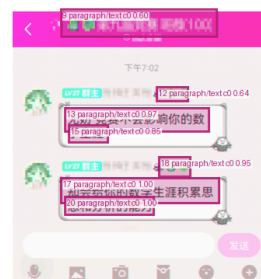
作者：清疏数学

组织：清疏大学

时间：July 7, 2024

版本：2.0

作者联系方式：QQ:937821418



本讲义将自动生成和 cctalk 用户绑定的 ID, 如果你的 ID 多次出现在公开环境, 我们会追究法律责任

仕。

1 paragraph/text c0 0.99

Problem Books in Mathematics

1 paragraph/text c0 0.99

Ovidiu Furdui

2 paragraph/text c0 1.00

3 paragraph/text c0 1.00

4 paragraph/text c0 1.00

5 paragraph/text c0 1.00

6 paragraph/text c0 1.00

7 paragraph/text c0 1.00

8 paragraph/text c0 0.98

Limits, Series, and Fractional Part Integrals

Problems in Mathematical Analysis



Springer

- 10
- 495
- 21,417
- Markdown 304
-
- 1.33
- 18.53

MathCraft OCR

- ONNX Runtime PyTorch
- manifest
- /
- OCR
-

MATHCRAFT_FORCE_ORT_CPU MATHCRAFT_HOME

Windows (永久生效):

1. Win+R → 输入 sysdm.cpl → 高级 → 环境变量
2. 在"用户变量"中新建, 变量名填入上面给出的名称, 值填入 1 或路径

Windows (仅当前终端):

在 PowerShell 中执行: `$env:MATHCRAFT_FORCE_ORT_CPU = "1"`

Linux / macOS (永久生效):

在 `~/.bashrc` 或 `~/.zshrc` 末尾添加:
`export MATHCRAFT_FORCE_ORT_CPU=1`
然后执行 `source ~/.bashrc`

Linux / macOS (仅当前终端):

`export MATHCRAFT_FORCE_ORT_CPU=1`

- MATHCRAFT_FORCE_ORT_CPU — 1 CPU GPU
- MATHCRAFT_HOME — MathCraft
- HTTP_PROXY / HTTPS_PROXY —
- MATHCRAFT_LOG_LEVEL — DEBUG / INFO / WARNING / ERROR
- LATEXSNIPPER_FORCE_LINUX_GRAPHICS_FALLBACKS — Linux Qt/WebEngine
- LATEXSNIPPER_DISABLE_LINUX_GRAPHICS_FALLBACKS — Linux Qt/GPU
- LATEXSNIPPER_SHOW_CONSOLE — Windows